Fig.	*Endpoint*	*Condition/Sample*	*Unit plotted*	*Mice/Gel (N)*	*Total cells (n)*
1d	Major axis diameter	Pre- vs. post-expansion	Per-cell	Gel: 3	13
1e	Expansion factor (1x PBS)	Pre- vs. post-expansion	Per-cell	Gel: 3	13
2c	Cell surface area	Naïve, Effector T cell	Per-cell	Naïve: 1; Effector: 4	Naïve: 10; Effector: 17
2d	Density	Naïve, Effector T cell	Per-cell	Naïve: 1; Effector: 4	Naïve: 10; Effector: 17
3m	Density	Effector T cell	Per-cell	4	17
3n	MV length, diameter	Effector T cell	Per-cell mean	4	17
3o	Patch density	TCR, CD2, LFA-1	Per-cell mean	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
3p	Patch diameter	TCR, CD2, LFA-1	Per-cell mean	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
4e	Patches per microvillus	TCR, CD2, LFA-1	Per-cell mean	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
4f	MV (%) containing each receptor	TCR, CD2, LFA-1	Per-cell	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
4g	Occupancy in MV region (voxel overlap)	TCR, CD2, LFA-1	Per-cell	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
4h	Intensity in MV relative to whole cell	TCR, CD2, LFA-1, membrane	Per-cell	TCR: 4; CD2: 4; LFA-1: 2; Membrane: 2	TCR: 14; CD2: 14; LFA-1: 10; Membrane: 10
4i	Distribution of distance ratio	TCR, CD2, LFA-1 to MV length	Per-patch	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
4j	Distribution of distance ratio	CD2 to TCR ($CD2^+TCR^+$ MV)	Per-MV	2	10
4k	Distribution of distance ratio	CD2 to LFA-1 ($CD2^+LFA\text{-}1^+$ MV)	Per-MV	2	10
4l	Distance ratio (scatter of 4j–k)	CD2-TCR; CD2-LFA-1	Per-MV	CD2-TCR: 2; CD2-LFA-1: 2	CD2-TCR: 10; CD2-LFA-1: 10
4m	Tip preference cell score	CD2-TCR; CD2-LFA-1; TCR-LFA-1	Per-cell score	CD2-TCR: 2; CD2-LFA-1: 2; TCR-LFA-1: 2	CD2-TCR: 12; CD2-LFA-1: 10; TCR-LFA-1: 12
5e	Colocalization with TCR (voxel overlap)	CD2, LFA-1 (whole cell)	Per-cell	CD2: 3; LFA-1: 2	CD2: 12; LFA-1: 8
5f	Colocalization with TCR (voxel overlap)	CD2, LFA-1 (MV region)	Per-cell	CD2: 3; LFA-1: 2	CD2: 12; LFA-1: 8
5g	Colocalization with TCR (voxel overlap)	CD2, LFA-1 (non-MV region)	Per-cell	CD2: 3; LFA-1: 2	CD2: 12; LFA-1: 8
6b	Patch density	TCR: 0 h, 1 h, 2 h	Per-cell mean	3 per time point	14 per time point
6c	Patches per microvillus	TCR: 0 h, 1 h, 2 h	Per-cell mean	3 per time point	14 per time point

6e	Patch density	CD2: 0 h, 1 h, 2 h	Per-cell mean	3 per time point	14 per time point
6f	Patches per microvillus	CD2: 0 h, 1 h, 2 h	Per-cell mean	3 per time point	14 per time point
6h	Patch density	LFA-1: 0 h, 1 h, 2 h	Per-cell mean	3 per time point	14 per time point
6i	Patches per microvillus	LFA-1: 0 h, 1 h, 2 h	Per-cell mean	3 per time point	14 per time point
6j	Percentage of MV types (heat map)	0 h, 1 h, 2 h	Mean percentage	0 h: 3; 1 h: 2; 2 h: 2	0 h: 12; 1 h: 8; 2 h: 10
6k	Percentage of MV types	0 h, 1 h, 2 h	Per-cell	0 h: 3; 1 h: 2; 2 h: 2	0 h: 12; 1 h: 8; 2 h: 10
6l	Tip preference cell score	0 h, 1 h, 2 h	Per-cell score	0 h: 3; 1 h: 2; 2 h: 2	0 h: 12; 1 h: 8; 2 h: 10
6m	Max distance from MV base	$CD2^+TCR^+$ MV: 0 h, 1 h, 2 h	Per-MV	0 h: 3; 1 h: 2; 2 h: 2	0 h: 12; 1 h: 8; 2 h: 10
6n	Distance ratio categories	0 h, 1 h, 2 h	Per-cell	0 h: 3; 1 h: 2; 2 h: 2	0 h: 12; 1 h: 8; 2 h: 10
7e	Intensity peak relative to distance	TCR-CD2	Single synapse	1 cell	1 synapse
7f	Intensity peak relative to distance	TCR-LFA-1	Single synapse	1 cell	1 synapse
S1b	Expansion factor (1x PBS)	Pre- vs. post-expansion	Per-cell	3 gels	Gel 1: 5; Gel 2: 4; Gel 3: 4
S1d	RMS measurement error vs. length	C1, C2, WGA	Per-cell mean	C1: 2 gels; C2: 3 gels; C3: 3 gels	C1: 6; C2: 8; C3: 8
S1f	Major axis diameter of nuclei	pre- vs. post-expansion (1x PBS)	Per-cell	Naïve: 1 gel; Effector: 1 gel	Naïve: 20; Effector: 20
S1h	Major axis diameter of nuclei	pre- vs. post-expansion (H_2O)	Per-cell	Jurkat: 1 gel; OT-I: 1 gel	Jurkat: 11; OT-I: 20
S2e	Occupancy in MV region (voxel overlap)	β-actin	Per-cell	1	4
S2f	MV (%) containing β-actin	β-actin	Per-cell	1	4
S3b	MV length, diameter (manual)	Single cell	Per-MV	1	1 cell (50 MV)
S3c	MV length, diameter (NanoMAP segmentation)	Single cell (same cell as S3b)	Per-MV	1	1 cell (278 MV)
S4d	Patch area	TCR, CD2, LFA-1	Per-cell mean	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
S4e	Patch volume	TCR, CD2, LFA-1	Per-cell mean	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
S4f	Number of voxels per patch	TCR, CD2, LFA-1	Per-cell mean	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
S4g	Patch sphericity	TCR, CD2, LFA-1	Per-cell mean	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18

S4h	Patch distance from MV base	TCR, CD2, LFA-1	Per-cell mean	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
S4i	Distance ratio (patch to MV length)	TCR, CD2, LFA-1	Per-cell mean	TCR: 6; CD2: 6; LFA-1: 4	TCR: 22; CD2: 24; LFA-1: 18
S5g	Relative intensity enrichment in MV	TCR, CD2 (single cell)	Per z-slice	1 cell	48 z-slices
S5h	Relative intensity enrichment in MV	CD2, TCR, LFA-1, CD45, membrane	Per-cell	CD2: 6; TCR: 6; LFA-1: 4; CD45: 2; Membrane: 6	CD2: 25; TCR: 22; LFA-1: 19; CD45: 12; Membrane: 33
S6e	Colocalization with CD2 (voxel overlap)	TCR, LFA-1 (whole cell)	Per-cell	TCR: 3; LFA-1: 2	TCR: 12; LFA-1: 9
S6f	Colocalization with CD2 (voxel overlap)	TCR, LFA-1 (MV region)	Per-cell	TCR: 3; LFA-1: 2	TCR: 12; LFA-1: 9
S6g	Colocalization with CD2 (voxel overlap)	TCR, LFA-1 (non-MV region)	Per-cell	TCR: 3; LFA-1: 2	TCR: 12; LFA-1: 9
S6h	Colocalization with LFA-1(voxel overlap)	CD2, TCR (whole cell)	Per-cell	CD2: 2; TCR: 2	CD2: 9; TCR: 8
S6i	Colocalization with LFA-1 (voxel overlap)	CD2, TCR (MV region)	Per-cell	CD2: 2; TCR: 2	CD2: 9; TCR: 8
S6j	Colocalization with LFA-1 (voxel overlap)	CD2, TCR (non-MV region)	Per-cell	CD2: 2; TCR: 2	CD2: 9; TCR: 8
S8b	Distribution of distance ratio (to MV length)	TCR: 0 h, 1 h, 2 h	Per-patch	3 per time point	14 per time point
S8d	Distribution of distance ratio (to MV length)	CD2: 0 h, 1 h, 2 h	Per-patch	3 per time point	14 per time point
S8f	Distribution of distance ratio (to MV length)	LFA-1: 0 h, 1 h, 2 h	Per-patch	3 per time point	14 per time point
S8g	MV density	0 h, 1 h, 2 h	Per-cell	4 per time point	0 h: 17; 1 h: 18; 2 h: 20
S8h	MV (%) containing each receptor	TCR, CD2, LFA-1: 0 h, 1 h, 2 h each	Per-cell	3 per time point	14 per time point
S8i	Occupancy in MV region (voxel overlap)	TCR, CD2, LFA-1: 0 h, 1 h, 2 h each	Per-cell	3 per time point	14 per time point
S8j	Patches in MV relative to whole cell	TCR, CD2, LFA-1: 0 h, 1 h, 2 h each	Per-cell	3 per time point	14 per time point
S10c	Patch count (%) relative to whole T cell	TCR-CD2: synapse and non-synapse	Per-cell	2	4

S10d	Volume (%) relative to whole T cell	TCR-CD2: synapse and non-synapse	Per-cell	2	4
S10e	Intensity sum (%) relative to whole T cell	TCR-CD2: synapse and non-synapse	Per-cell	2	4
S10g	Patch count (%) relative to whole conjugate	TCR-CD2: T cell and DC membrane	Per-cell	2	5
S10h	Voxel number (%) relative to whole conjugate	TCR-CD2: T cell and DC membrane	Per-cell	2	5
S10i	Intensity sum (%) relative to whole conjugate	TCR-CD2: T cell and DC membrane	Per-cell	2	5
S11d	Patch density	TCR: 0 h, 1 h, 2 h (MC38 co-incubation)	Per-cell	1	3 per time point
S11e	Patch density	CD2: 0 h, 1 h, 2 h (MC38 co-incubation)	Per-cell	1	3 per time point
S11f	Patch density	LFA-1: 0 h, 1 h, 2 h (MC38 co-incubation)	Per-cell	1	3 per time point
S11g	Occupancy in MV region (voxel overlap)	TCR: 0 h, 1 h, 2 h (MC38 co-incubation)	Per-cell	1	3 per time point
S11h	Occupancy in MV region (voxel overlap)	CD2: 0 h, 1 h, 2 h (MC38 co-incubation)	Per-cell	1	3 per time point
S11i	Occupancy in MV region (voxel overlap)	LFA-1: 0 h, 1 h, 2 h (MC38 co-incubation)	Per-cell	1	3 per time point